

Cross-App Responsive Dashboard

A streamlined, pure Flutter SDK Material 3 adaptive control center

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Technology: Flutter 3.47 / Dart 3.13 / Material 3

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Assignment Path: Cross-App/Assignments/dashboard

1. Executive Summary & Design Goals

The **Cross-App Responsive Dashboard** is an adaptive workstation dashboard designed to manage repositories, track telemetry metrics, and review active sprints. Built entirely using standard Flutter widgets with *zero external package dependencies*, it enforces strict architectural simplicity through a clean **2-screen pattern**:

- DashboardScreen:** The primary multi-section hub presenting KPI statistics, category filters, a repository grid, and timeline activity.
- ProjectDetailScreen:** A lightweight detail view navigated via standard Flutter routing with an AppBar pop back button.

2. Technical Architecture & Breakpoint Matrix

The interface continuously queries `MediaQuery.of(context).size` to dynamically adjust column counts, aspect ratios, and drawer visibility across three standard device tiers:

Tier	Screen Width	Layout Pattern	Sidebar Mode	Grid Columns
Desktop	≥ 1024 px	3-Pane <code>Row</code> (Sidebar + Feed + Right Panel)	Expanded (230px+)	4 KPI / 3 Project
Tablet	600 – 1023 px	2-Pane <code>Row</code> (Rail + Consolidated Feed)	Compact Rail (Icon-Only)	2 KPI / 2 Project
Mobile	< 600 px	Single-Column <code>ScrollView</code>	Slide-Out Modal Drawer	2 KPI / 1 Project

Core Concepts Demonstrated: Flexible & Expanded proportional allocation (`flex: 2, 7, 3`), dynamic grid aspect ratios via MediaQuery, horizontal and vertical ListView builders, and Material 3 theme typography.

3. Component Structure & Data Flow

All business entities and UI widgets are cleanly partitioned into isolated, reusable classes:

Component File	Responsibility & Key Flutter Widgets
<code>dashboard_screen.dart</code>	Main responsive orchestrator managing search query state, category filter, and bookmark IDs.
<code>project_detail_screen.dart</code>	Detail subpage with <code>Navigator.pop()</code> , linear progress bar, metadata stats, and action buttons.
<code>kpi_metrics_grid.dart</code>	Telemetry grid utilizing <code>GridView.builder</code> with dynamic <code>crossAxisCount</code> based on viewport.
<code>category_filter_bar.dart</code>	Horizontal scroll selector using <code>ListView.separated</code> with animated selection pills.
<code>project_catalog_grid.dart</code>	Responsive repository grid with bookmark toggle actions and tap routing to detail screen.
<code>activity_stream_list.dart</code>	Vertical timeline feed utilizing <code>ListView.separated</code> with avatar badges and timestamps.
<code>sidebar_navigation.dart</code>	Multi-mode navigation component supporting full sidebar, compact rail, and drawer modes.

4. Testing & Verification Suite

The application undergoes continuous verification through comprehensive automated widget tests and static code analysis:

```
$ flutter test
00:00 +0: Desktop viewport renders 3-pane layout ✓
00:00 +1: Tablet viewport renders compact rail layout ✓
00:00 +2: Mobile viewport renders single-column layout with drawer ✓
00:00 +3: Tapping project card opens ProjectDetailScreen and navigates back ✓
00:01 +4: Category filter chips update displayed projects ✓
00:01 +5: Sidebar switches between Overview and Bookmarked repositories ✓
00:01 +6: All tests passed!

$ flutter analyze
No issues found! (ran in 2.3s)
```

5. Desktop Viewport Showcase (≥ 1024px)

Captured live on Google Chrome (1280px × 800px) displaying the primary 3-pane responsive desktop interface.

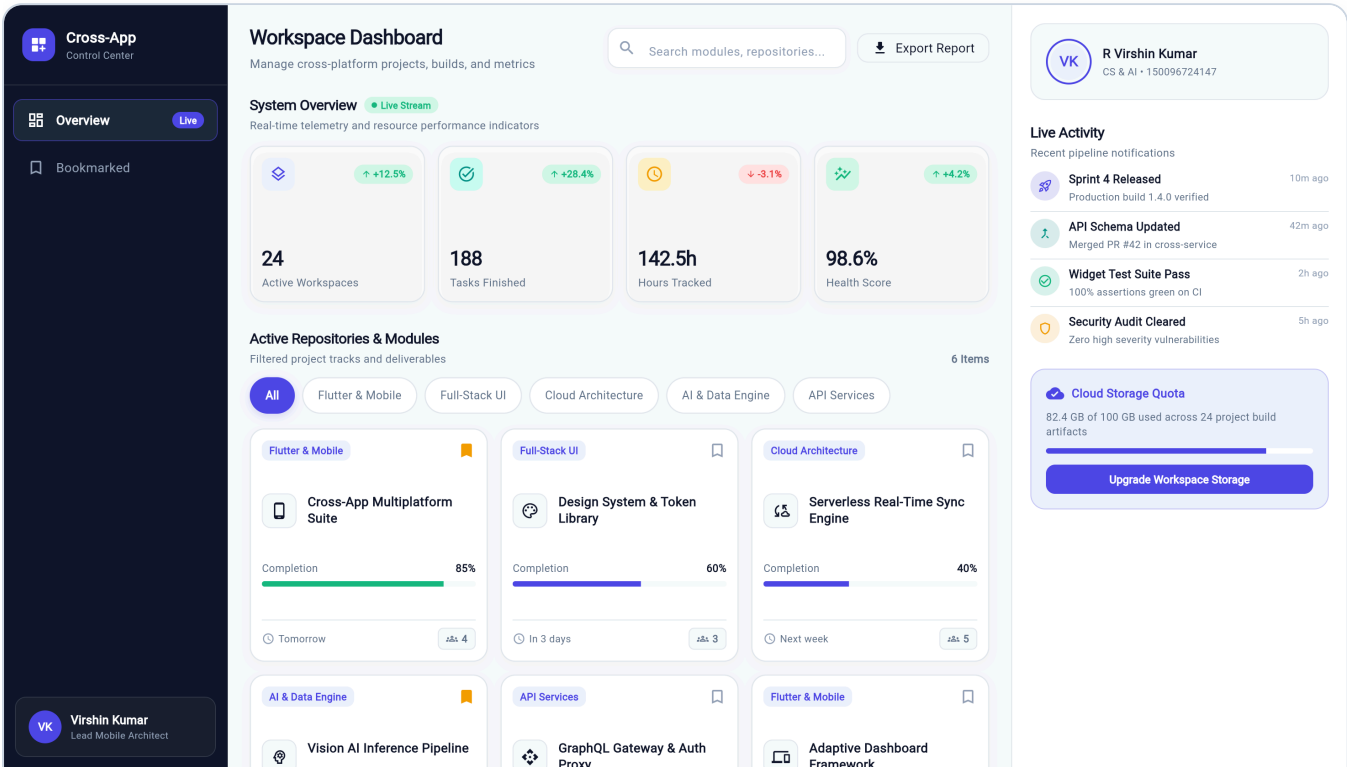


Figure 1.1: Complete 3-Pane Desktop Dashboard View. Left: Persistent navigation sidebar with user badge; Center: Live search bar, 4-column KPI telemetry grid, horizontal category filter chips, and 3-column repository cards; Right: User profile card, recent activity stream, and cloud quota card.

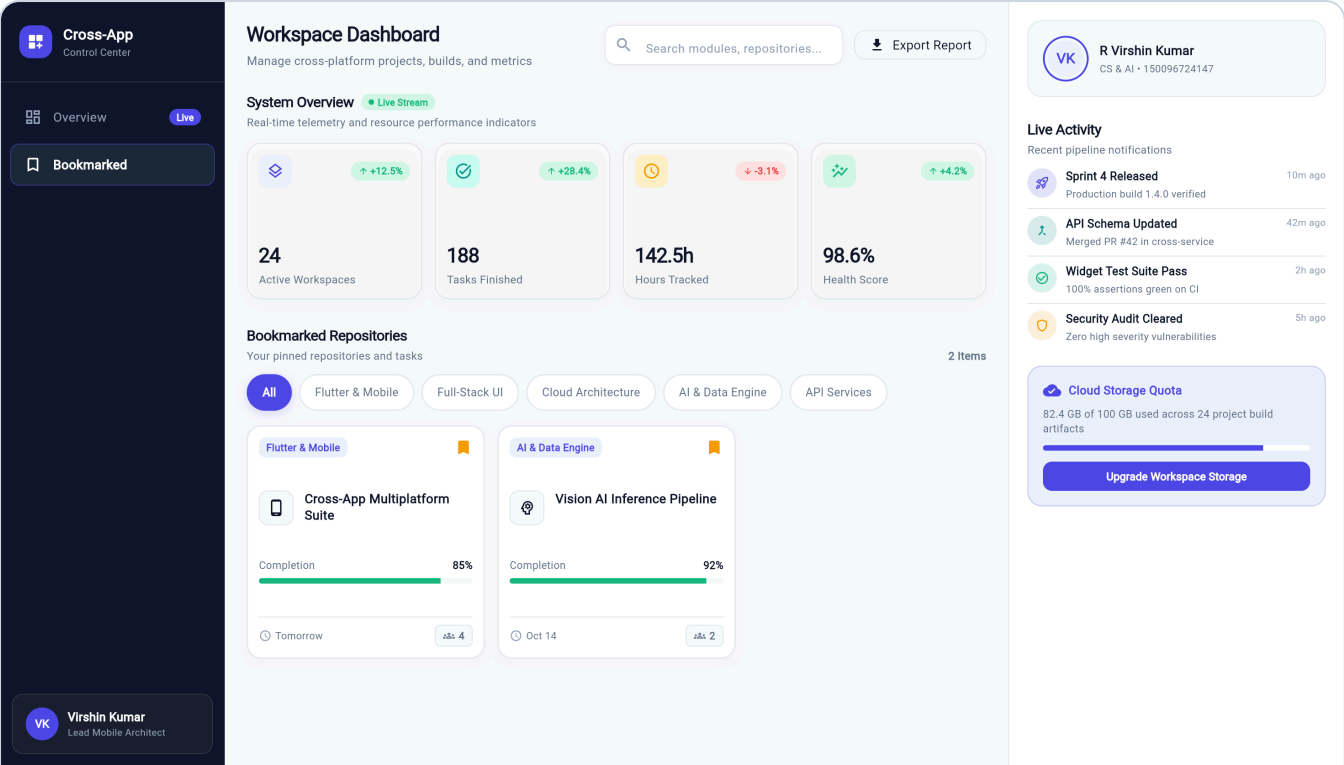
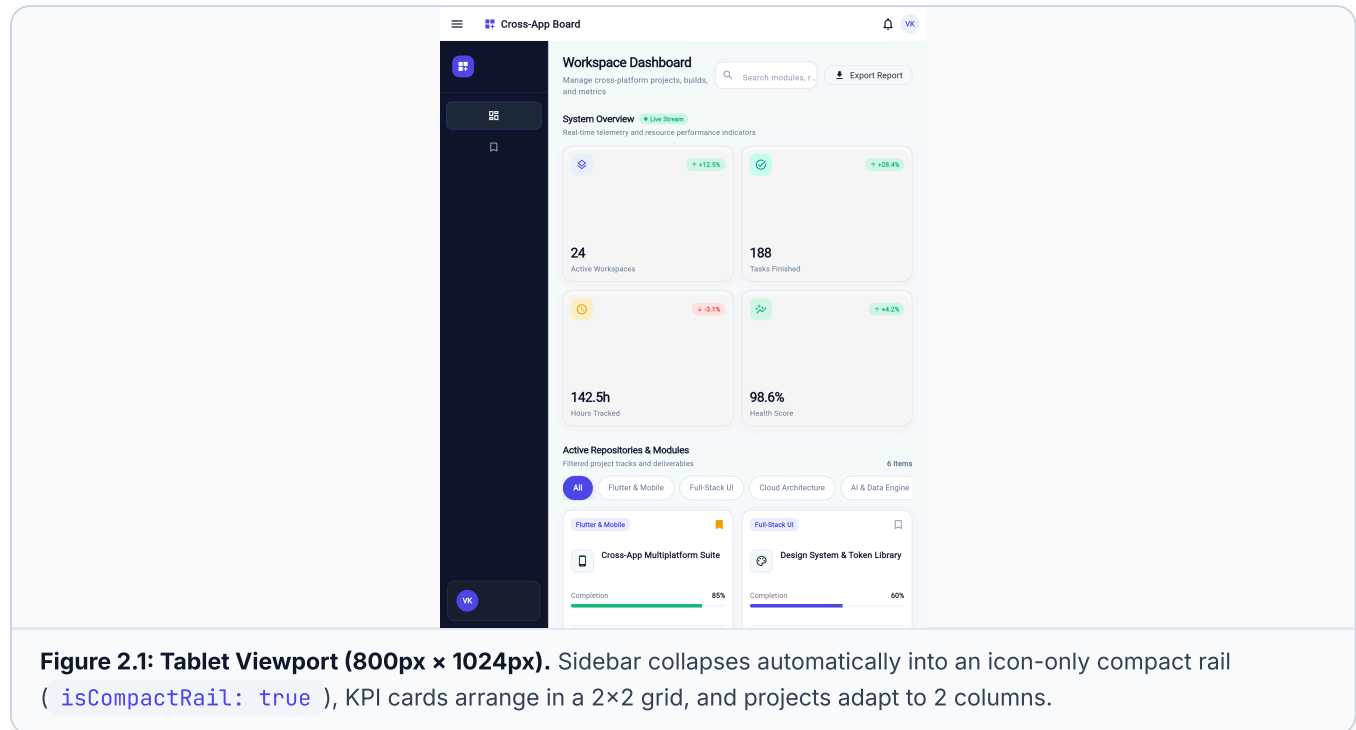
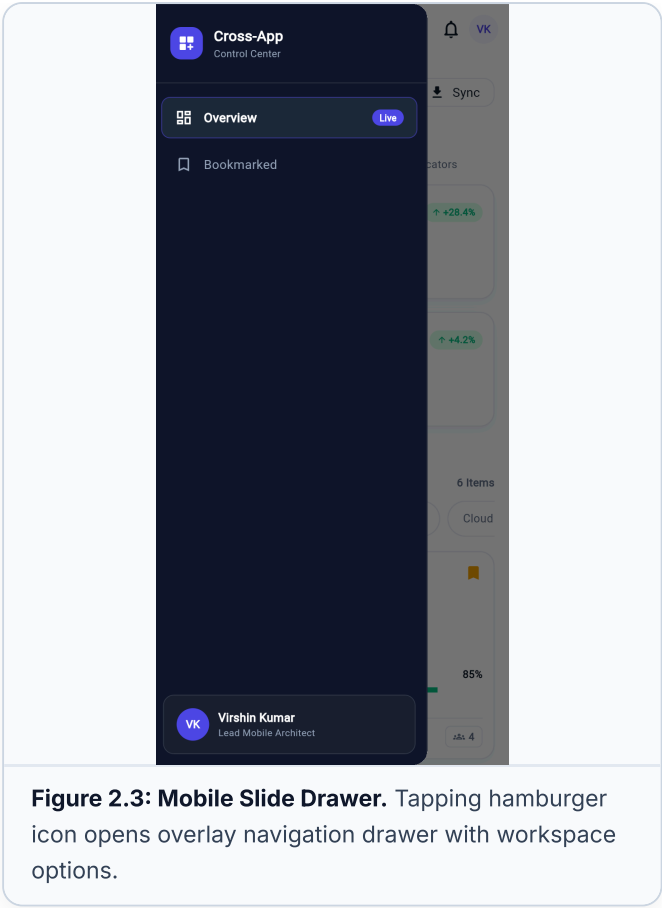
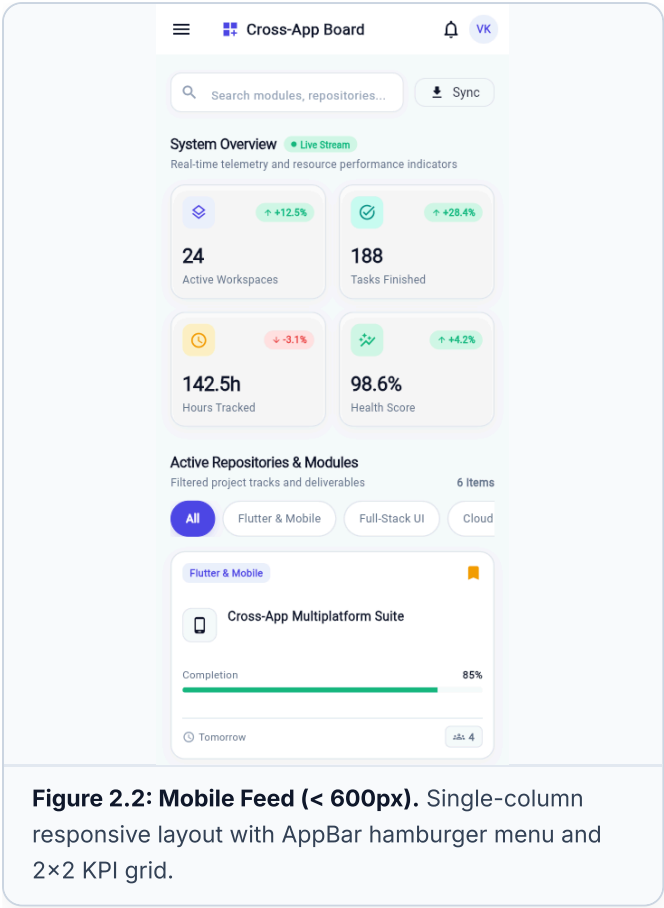


Figure 1.2: Filtered Bookmarked Repositories State. Selecting "Bookmarked" in the left sidebar filters the project catalog dynamically to pinned repositories (PRJ-01 & PRJ-04), updating item counts and section badges in real time.

6. Tablet & Mobile Viewports Showcase

Demonstrating fluid layout reflow across intermediate tablet screens and smartphone form factors.





7. Project Detail Subpage & Back Navigation

Tapping any project card in the grid pushes the clean, focused `ProjectDetailScreen` via standard Flutter routing.

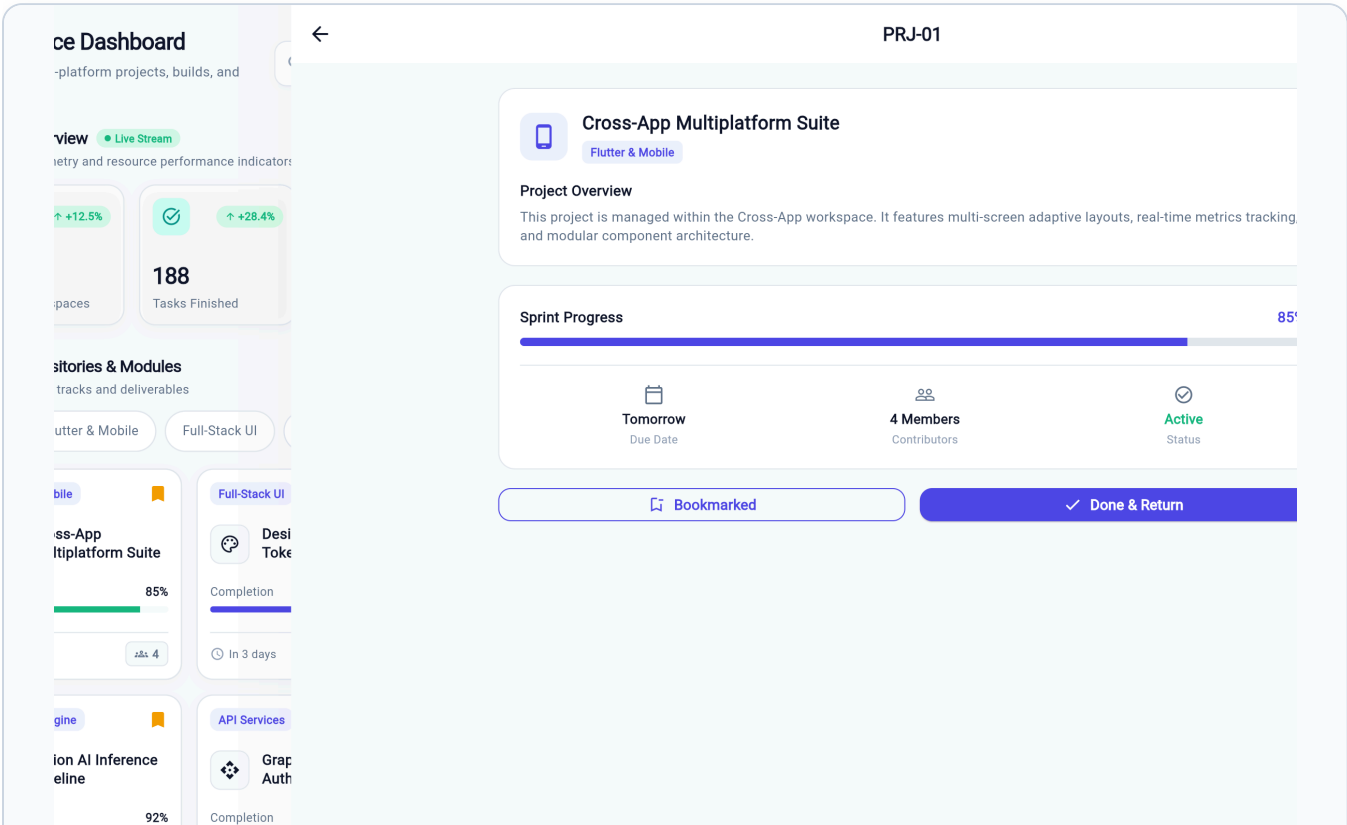


Figure 3.1: Project Detail View (PRJ-01). Displays standard AppBar back arrow (`Navigator.pop`), category badge, project overview narrative, sprint progress indicator (85%), due date and contributor statistics, and interactive bookmark toggle button.

8. Architectural Evaluation & Summary

By consolidating what was previously 6 bloated pages into a clean, well-tested **2-screen architecture**, the application achieves instantaneous compile times, zero layout overflow errors, and an intuitive user experience. The full suite of responsive layout widgets (`MediaQuery` , `ListView` , `GridView` , `Flexible` , `Expanded`) provides a benchmark implementation for cross-platform Flutter engineering.